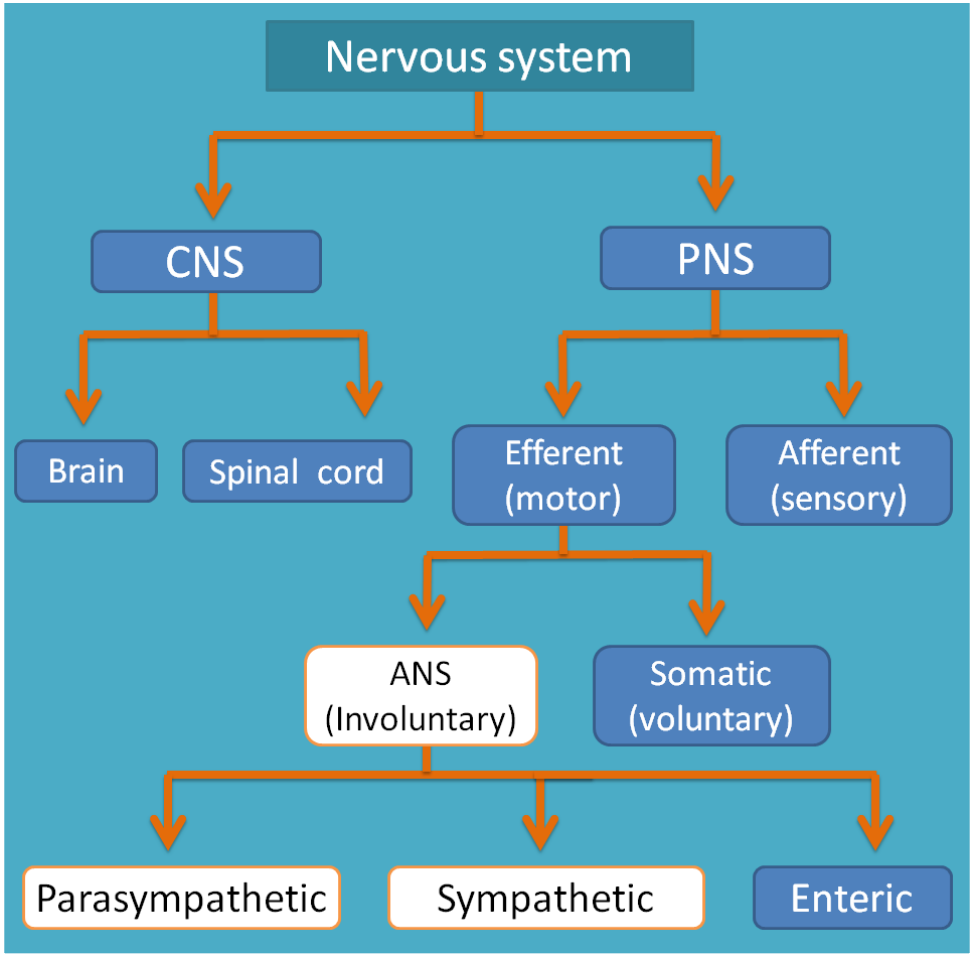
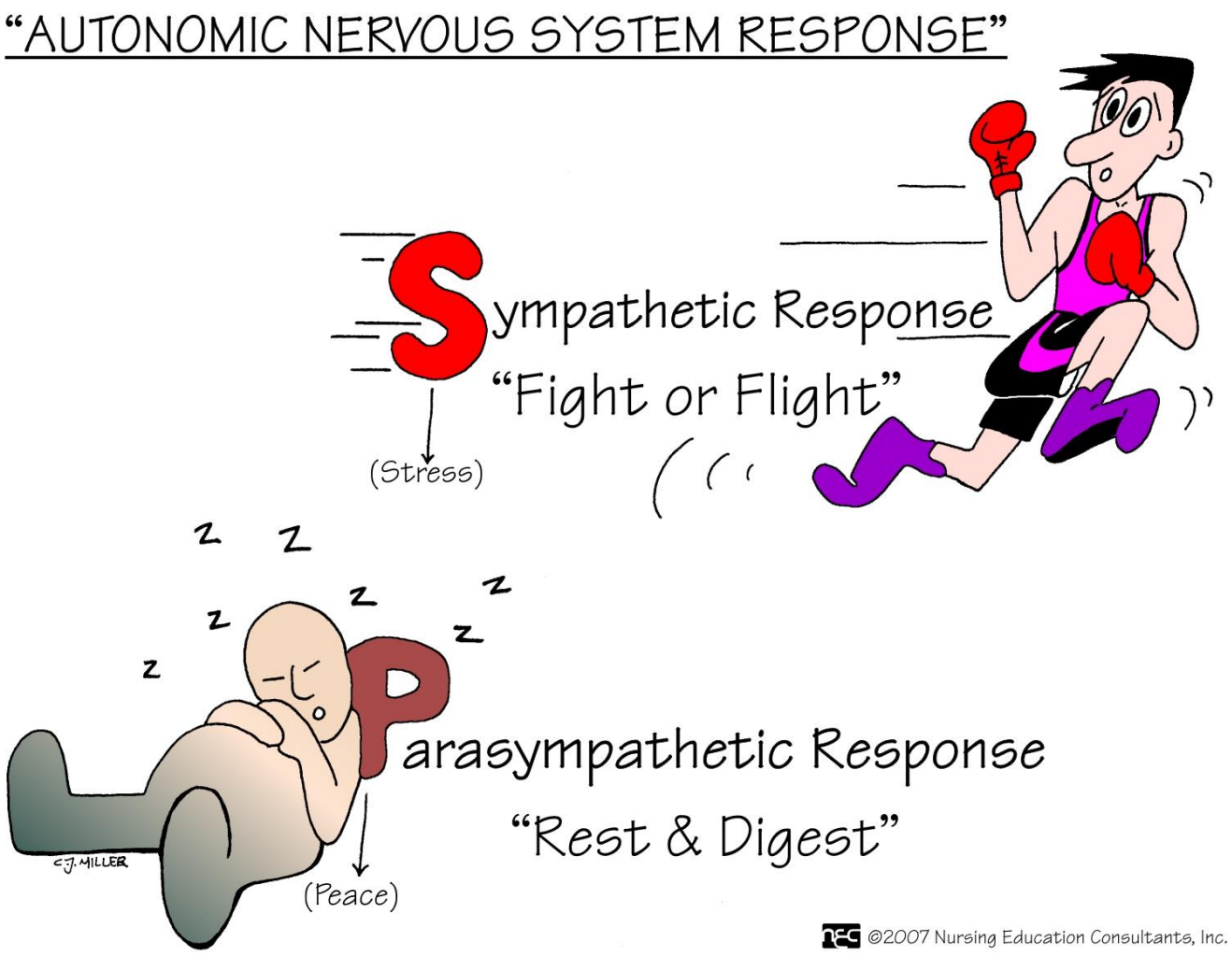
Unit 2
Autonomic Nervous System Pharmacology

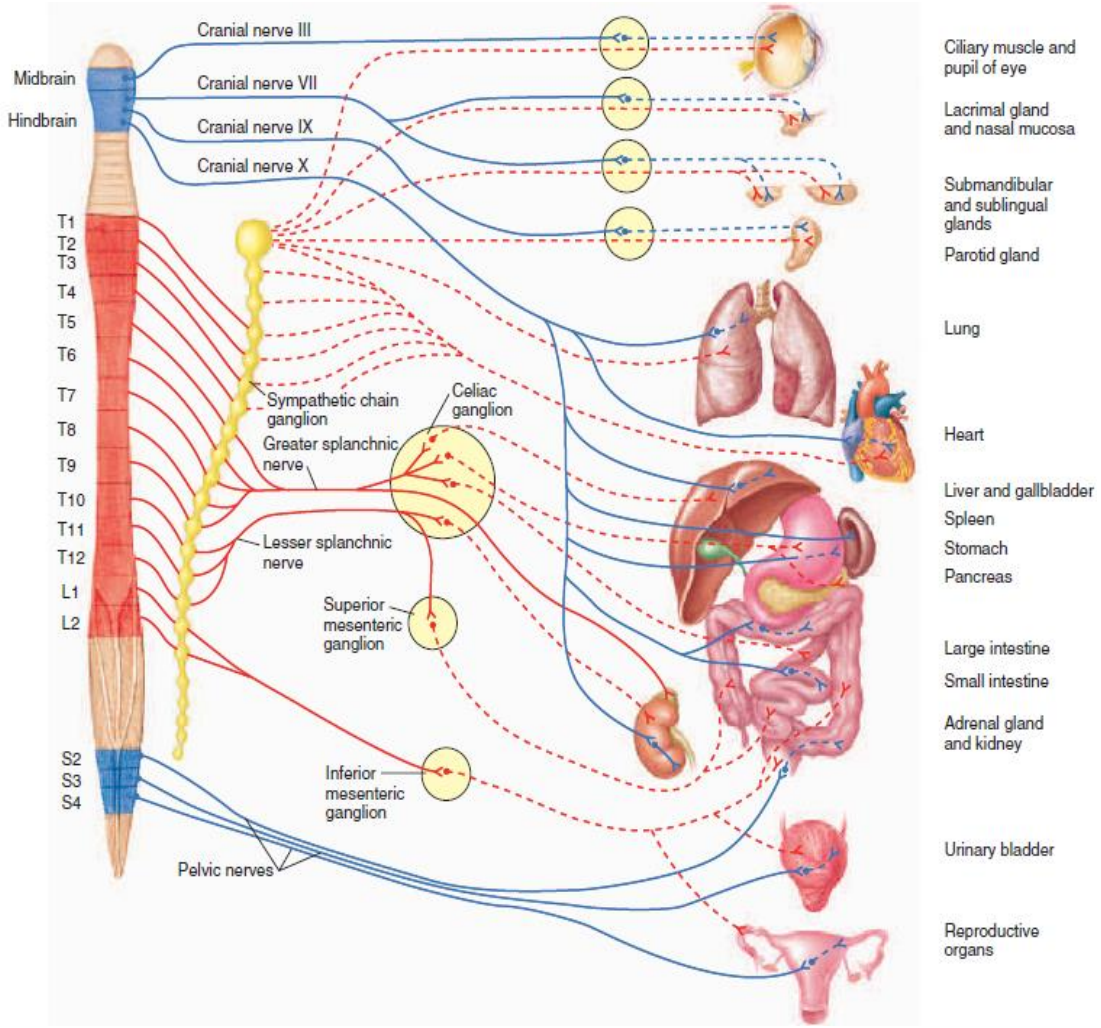
Physiology of nervous system



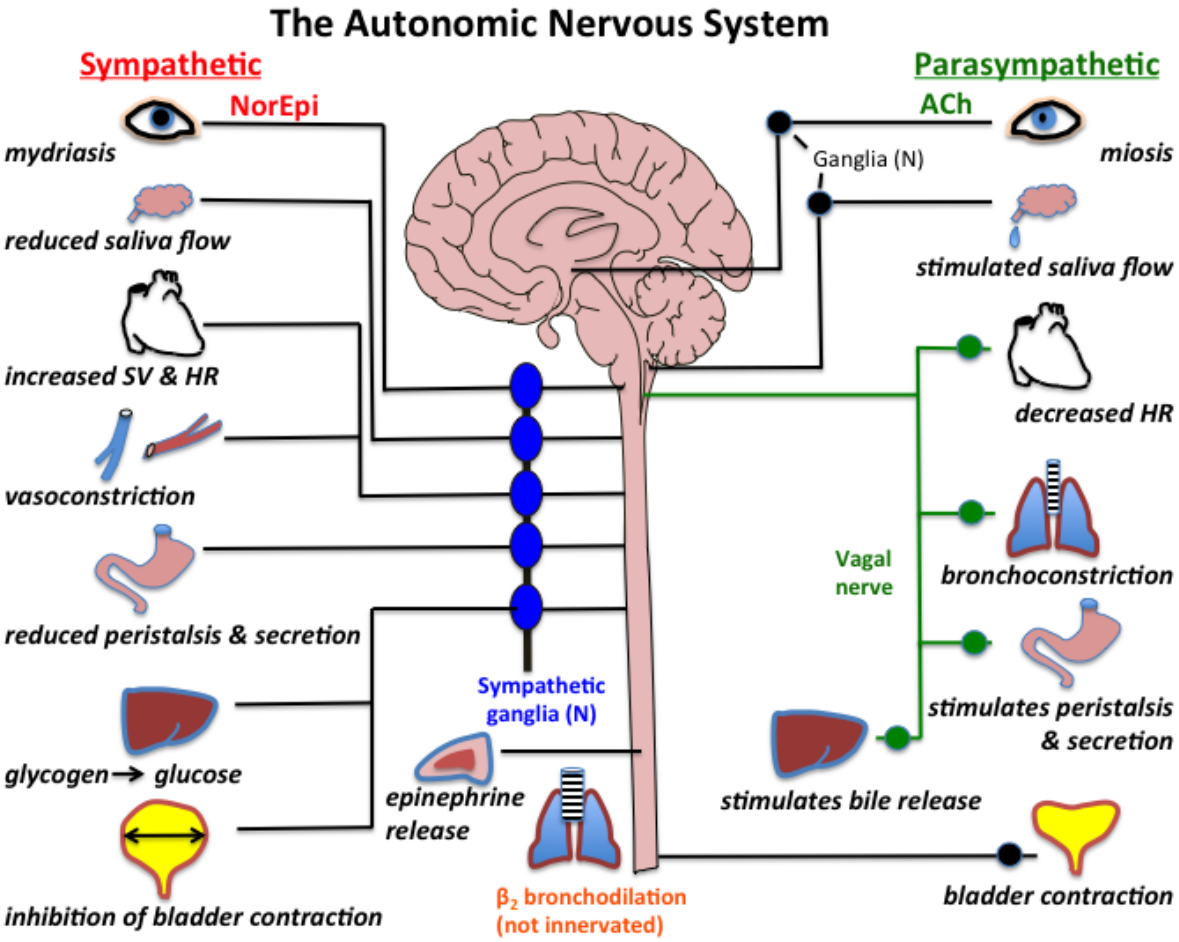
cont'd …


"AUTONOMIC NERVOUS SYSTEM RESPONSE"
Sympathetic Response
"Fight or Flight"
(Stress)
Parasympathetic Response
"Rest & Digest"
(Peace)
©J. MILLER
©2007 Nursing Education Consultants, Inc.

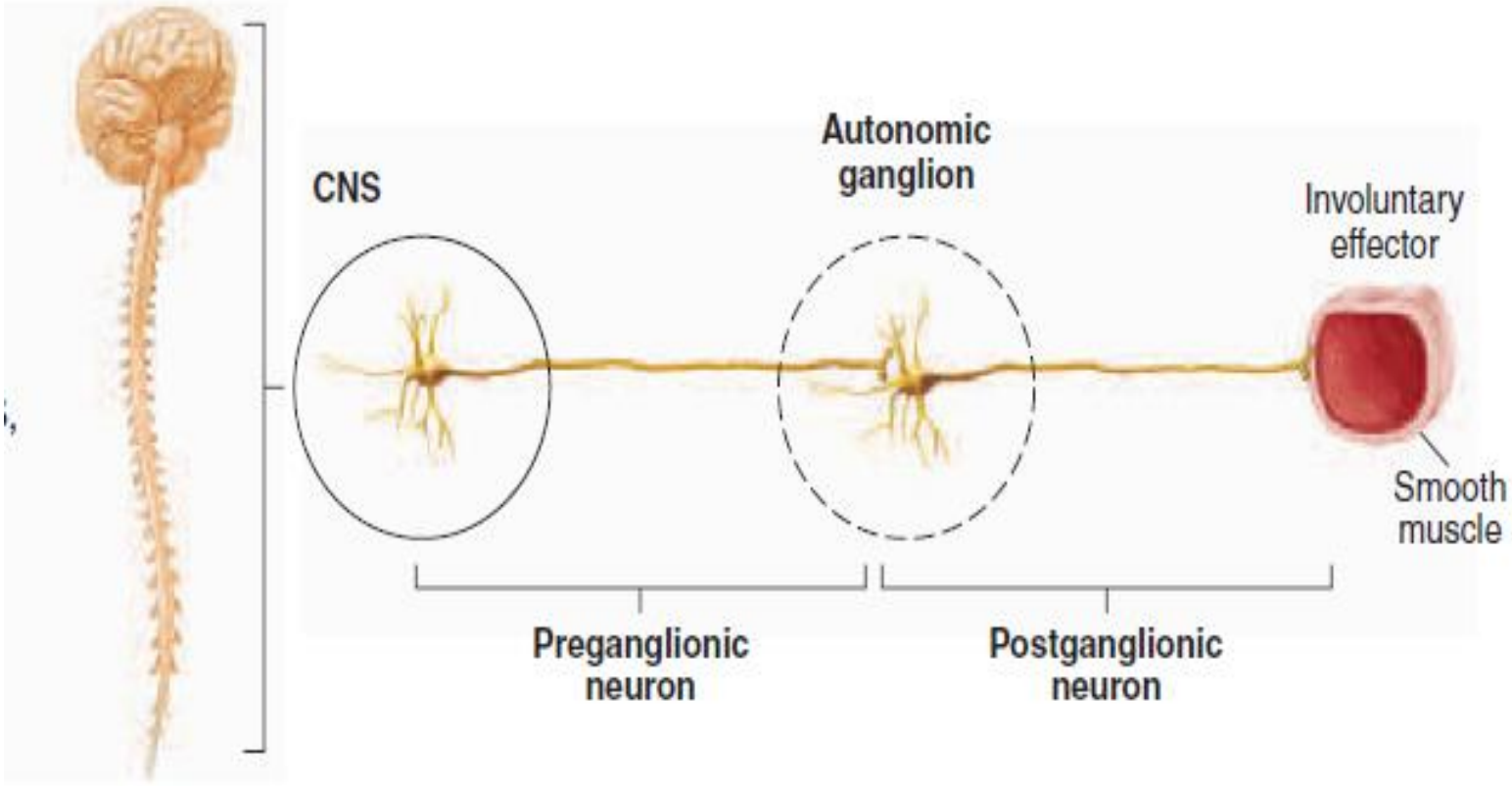
Origin of PSNS and SNS



Effect of ANS stimulation

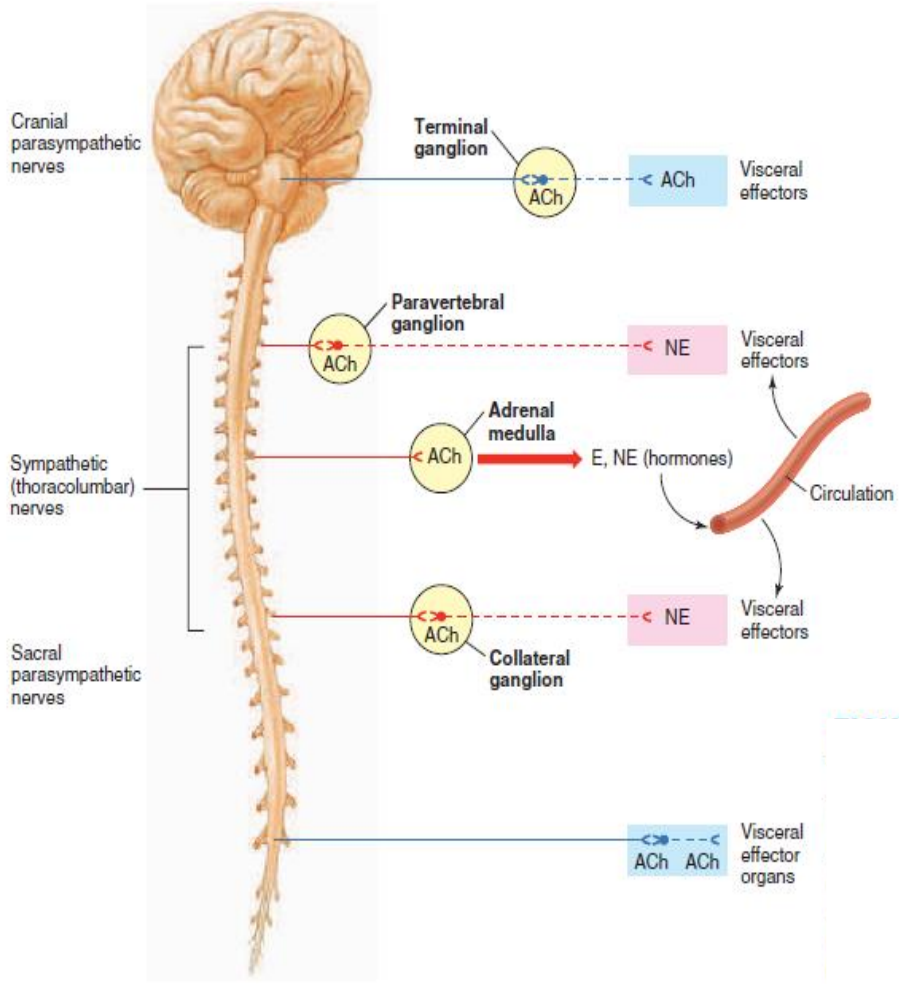


Signal transmission



Neurotransmitters in ANS

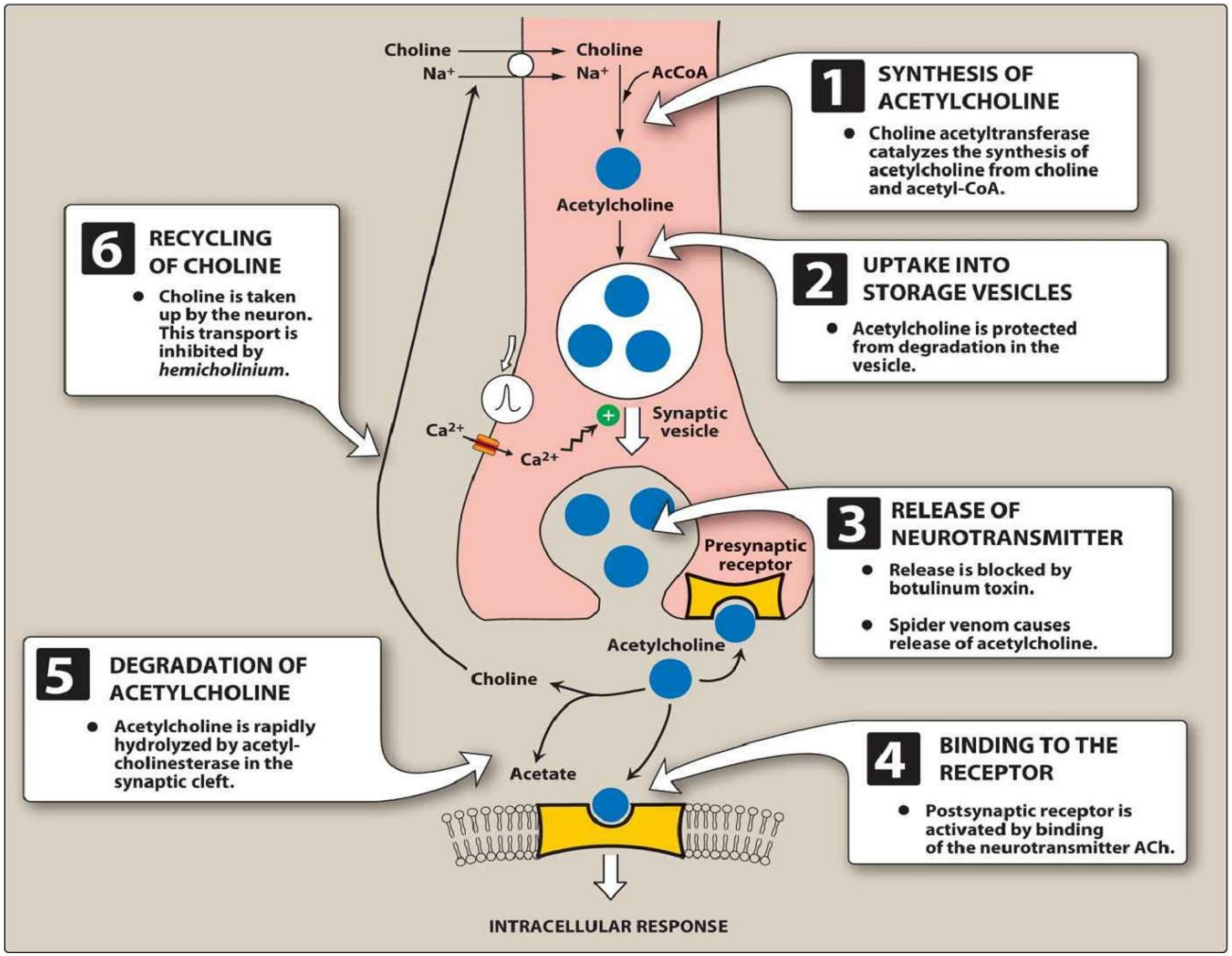
- Acetylcholine
- Norepinephrine



ANS drugs

- ✓ Cholinergic drugs
  - Agonists
  - Antagonists
- ✓ Adrenergic drugs
  - Agonists
  - Antagonists

Neurotransmission at cholinergic neurons



Cholinergic receptors

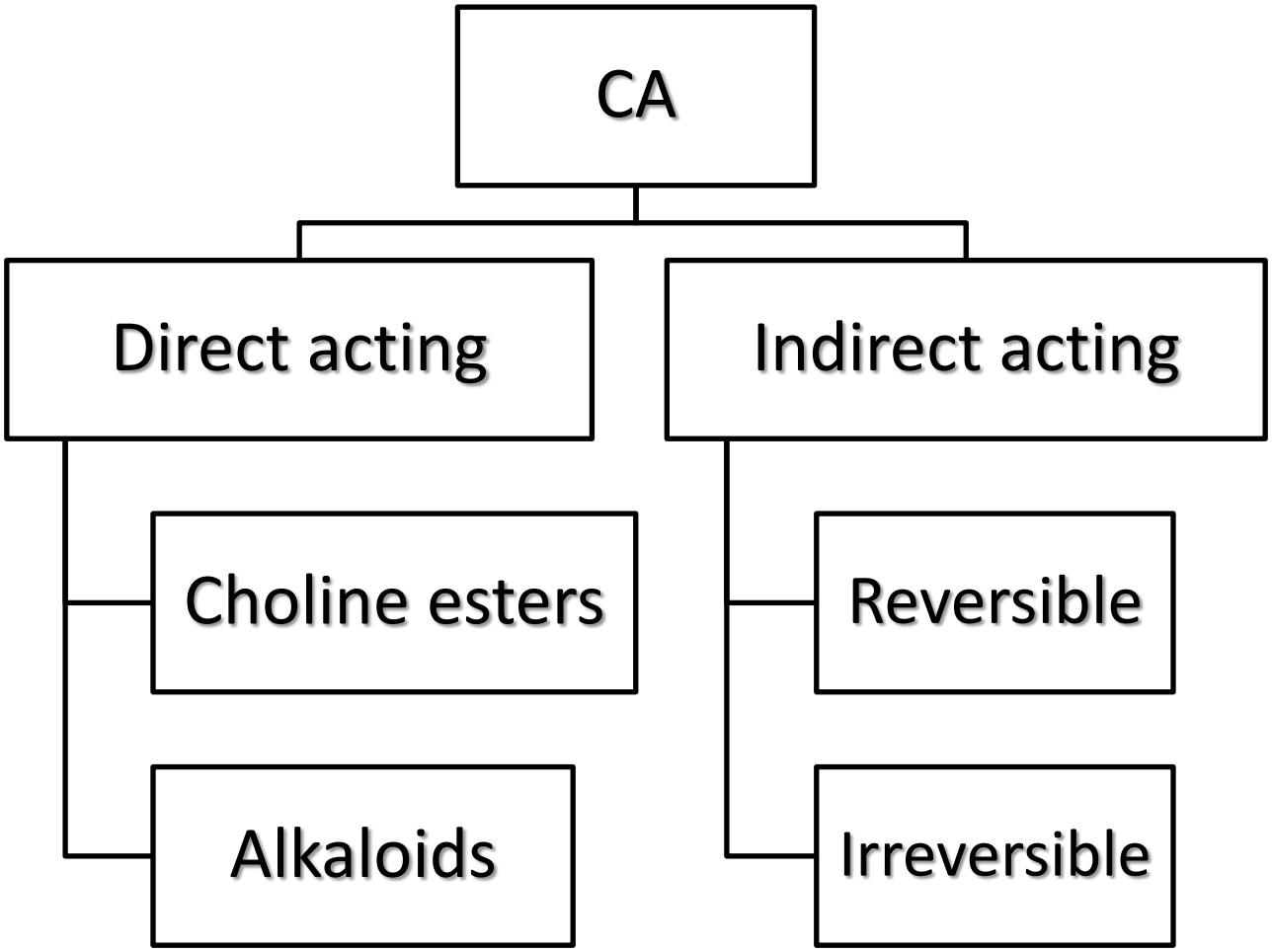
Muscarinic

- M_1 – CNS
- M_2 – cardiac cells, smooth muscle (lung, stomach)
- M_3 – bladder, exocrine glands, smooth muscle

Nicotinic

- N_n – CNS, adrenal medulla, autonomic ganglia
- N_m – neuromuscular junction in skeletal muscle

Cholinergic agonists (CA)



Direct vs. Indirect

- Direct acting agonists bind directly to acetylcholine receptors to produce a biological effect.

- Indirect acting agonists cannot bind directly to acetylcholine receptors but prevent hydrolysis of endogenous acetylcholine.

Direct acting agents

- ✓ Acetylcholine – prototype
- ✓ Bethanechol
- ✓ Carbachol
- ✓ Methacholine
- ✓ Nicotine
- ✓ Pilocarpine

Mnemonic - ABCMNoP

Pharmacokinetics

- Choline esters (ABCM) – insoluble in lipids
  - Poor absorption and distribution to CNS
  - All are hydrolyzed in GIT, thus orally less active
  - Differ in their susceptibility to hydrolysis by cholinesterases

- Alkaloids (NP) – lipid soluble
  - Well absorbed from most sites of administration.
  - Acidification of urine accelerates their clearance.

cont'd …

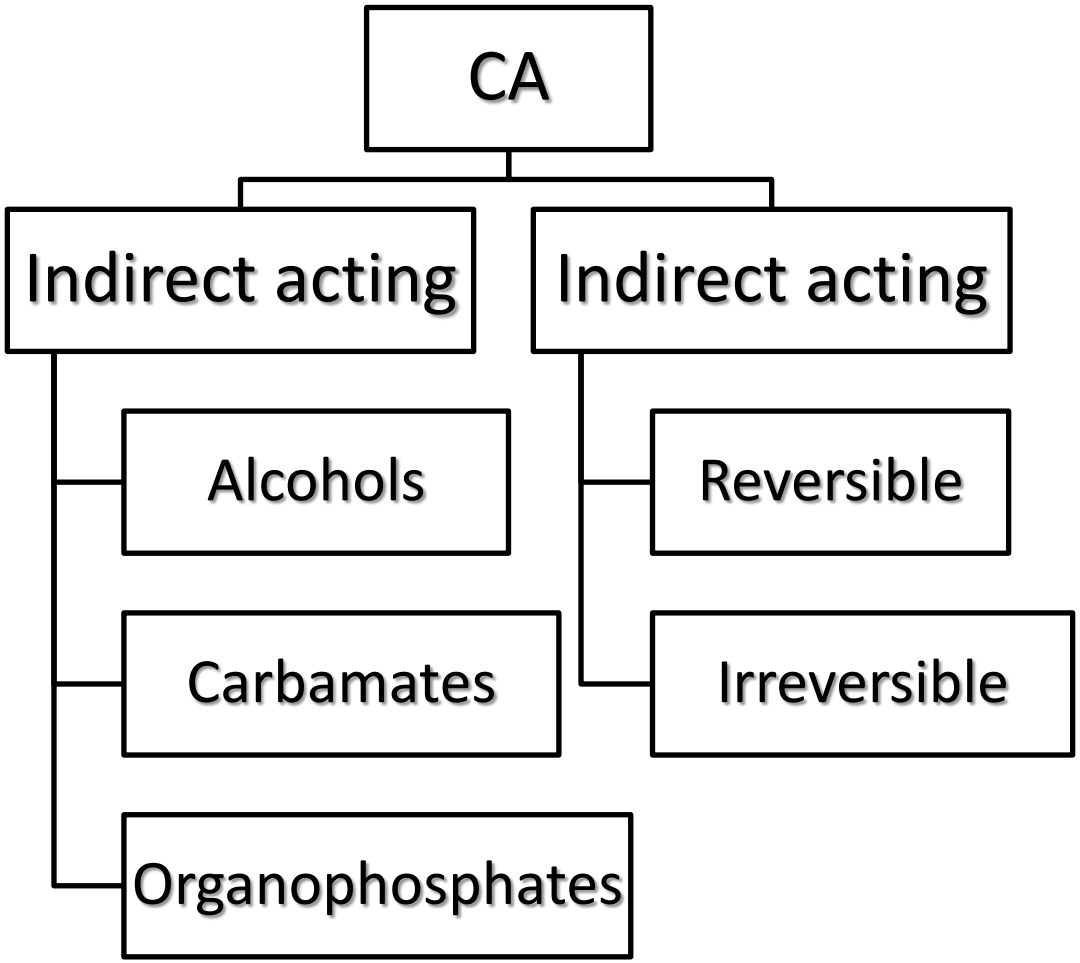
Choline esters differ markedly in their susceptibility to hydrolysis by cholinesterase.

Choline Ester	Susceptibility to Cholinesterase	Muscarinic Action	Nicotinic Action
Acetylcholine chloride	++++	+++	+++
Methacholine chloride	+	++++	None
Carbachol chloride	Negligible	++	+++
Bethanechol chloride	Negligible	++	None

Organ	Response
Eye	
Sphincter muscle of iris	Contraction (miosis)
Ciliary muscle	Contraction for near vision (accommodation)
Heart	
Sinoatrial node	Decrease in rate (negative chronotropy)
Atria	Decrease in contractile strength (negative inotropy). Decrease in refractory period
Atrioventricular node	Decrease in conduction velocity (negative dromotropy). Increase in refractory period
Ventricles	Small decrease in contractile strength
Blood vessels	
Arteries, veins	Dilation (via EDRF). Constriction (high-dose direct effect)
Lung	
Bronchial muscle	Contraction (bronchoconstriction)
Bronchial glands	Secretion
Gastrointestinal tract	
Motility	Increase
Sphincters	Relaxation
Secretion	Stimulation
Urinary bladder	
Detrusor	Contraction
Trigone and sphincter	Relaxation
Glands	
Sweat, salivary, lacrimal, nasopharyngeal	Secretion

Organ system effects

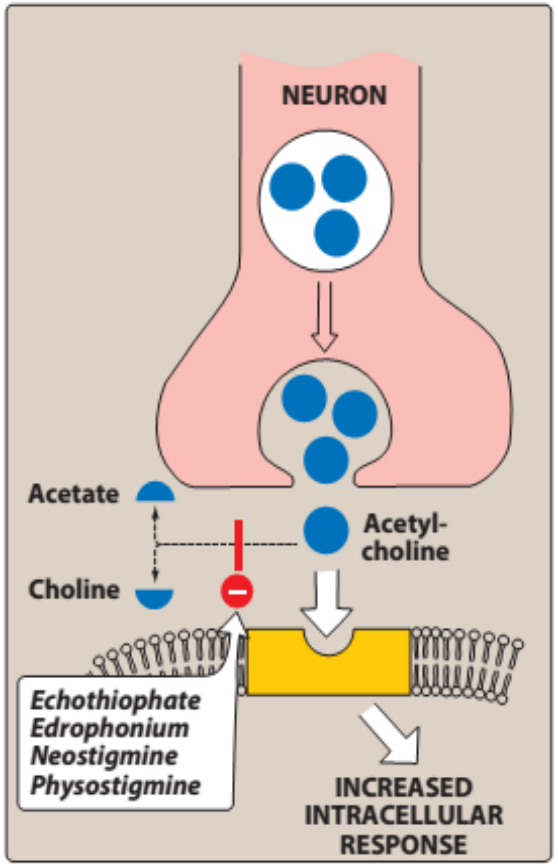
cont'd …


CA
Indirect acting
Indirect acting
Alcohols
Carbamates
Organophosphates
Reversible
Irreversible

Indirect acting cholinergic agonists

- AChE is an enzyme that cleaves ACh to acetate and choline.

- Inhibitors of AChE provide a cholinergic action by preventing the degradation of acetylcholine.



Indirect acting agents

- ✓ Edrophonium – prototype short acting agent
- ✓ Neostigmine
- ✓ Physostigmine
- ✓ Pyridostigmine
- ✓ Echothiopate – irreversible

Mnemonic – 2(PEN)

Cont'd…

Group, Drug	Uses	Approximate Duration of Action
Alcohols		
Edrophonium	Myasthenia gravis, ileus, arrhythmias	5–15 minutes
Carbamates and related agents		
Neostigmine	Myasthenia gravis, ileus	0.5–4 hours
Pyridostigmine	Myasthenia gravis	4–6 hours
Physostigmine	For anticholinergic poisoning	0.5–2 hours
Organophosphates		
Echothiophate	Glaucoma	100 hours

Basic pharmacology and PK

- The following are lipid insoluble
  - Edrophonium
  - Neostigmine
  - Pyridostigmine

→ Absorption from the conjunctiva, skin, gut and lungs is poor. Distribution into the CNS is negligible.

- Physostigmine is lipid soluble

- Many of the organophosphate agents are highly lipid soluble liquids except echothiophate.

Organ system effects

- Most prominent effects are on CV and GI systems, the eye, and skeletal NMJ.
- Eye, respiratory tract, GIT, GUT
  - Similar to direct acting agents
- CVS (net effect)
  - Modest bradycardia, fall in CO, increased VR, increased BP
- NMJ
  - Therapeutic dose – increased strength of contraction in muscles weakened by NMJ blockers.

Clinical pharmacology

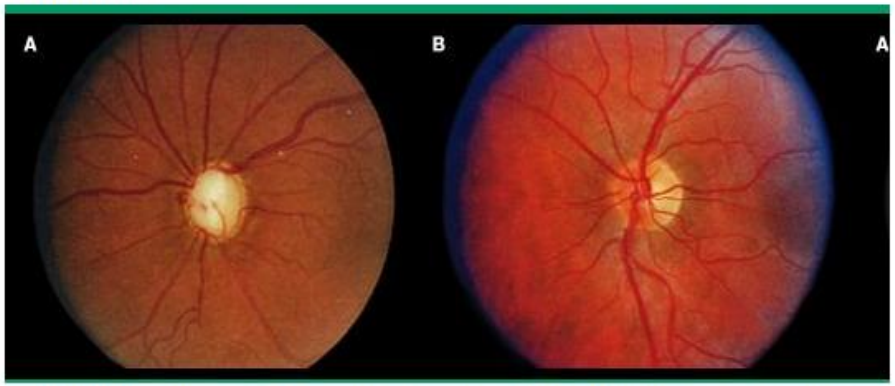
- The major therapeutic uses of the cholinomimetics are to treat diseases of the
  - Eye (glaucoma)
  - Gastrointestinal and urinary tracts (postoperative atony)
  - Neuromuscular junction (myasthenia gravis)

- Cholinesterase inhibitors are occasionally used in the treatment of atropine overdose and very rarely in the therapy of certain atrial arrhythmias.

1. Eye : Glaucoma

- Glaucoma is defined as an optic neuropathy.
- It is typically associated with elevated intra-ocular pressure (IOP).
- Leads to blindness if left untreated.

Glaucoma

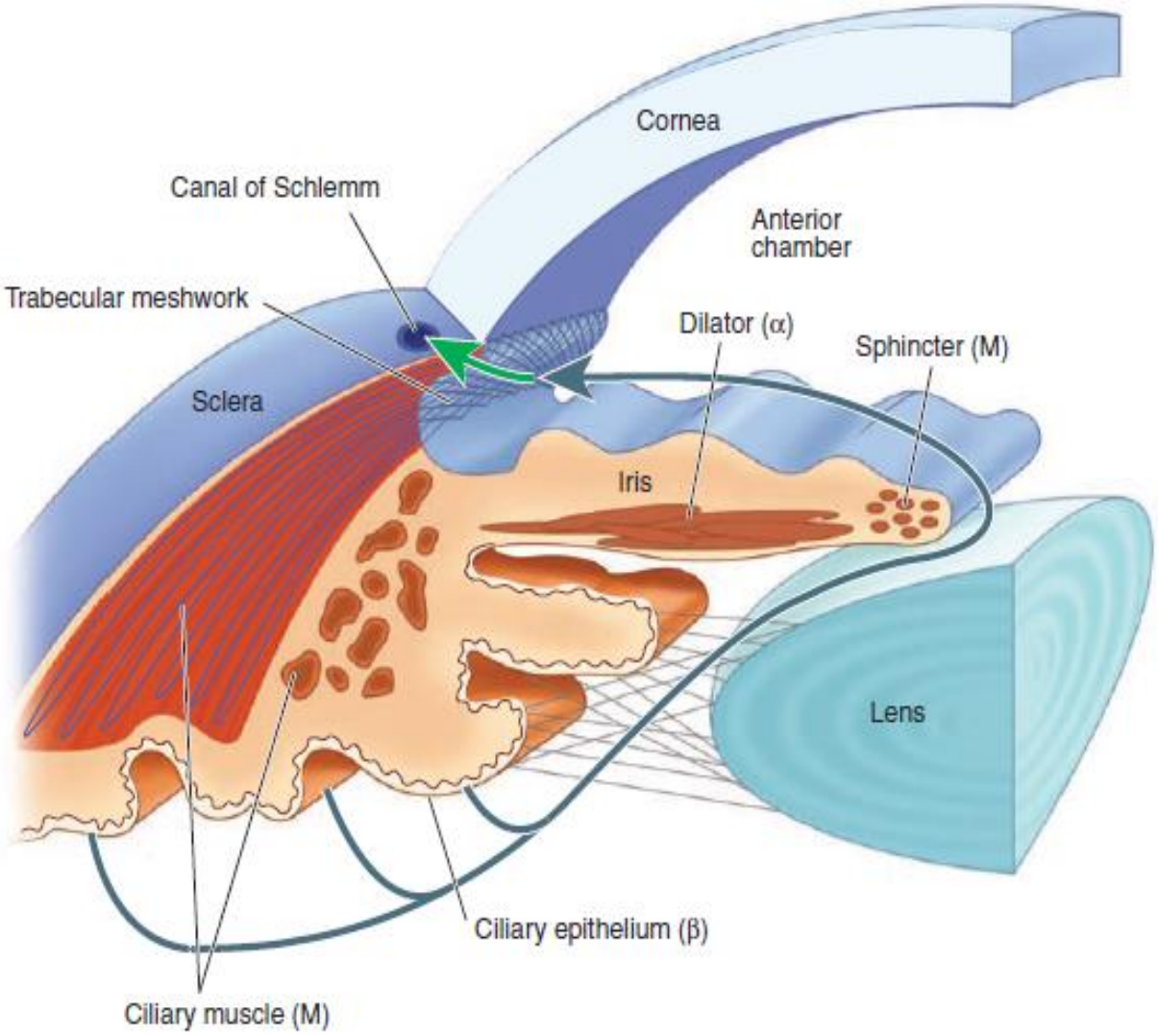


A) Extensive cupping of the optic disk typical of glaucoma. B) Compared to a normal optic disk.
Courtesy of the W.K. Kellogg Eye Center, University of Michigan.

Pharmacology of the eye

- Eye is an organ with multiple autonomic functions, controlled by several autonomic receptors.
- The anterior chamber is the site of several autonomic effector tissues.
- These tissues include 3 muscles :-
  1. Pupillary dilator
  2. Pupillary constrictor
  3. Ciliary muscle
- Secretory epithelium of the ciliary body.

cont'd …


Cornea
Canal of Schlemm
Anterior chamber
Trabecular meshwork
Dilator (α)
Sphincter (M)
Sclera
Iris
Lens
Ciliary epithelium (β)
Ciliary muscle (M)

cont'd …

- Muscarinic stimulants and cholinesterase inhibitors reduce IOP by :-
  - Causing contraction of the ciliary body

- Agents to be used include :-
  - Carbachol
  - Pilocarpine
  - Physostigmine
  - Ecothiophate

2. Gastrointestinal tract

- Cholinomimetic drugs may be helpful in clinical disorders that involve depression of smooth muscle activity without obstruction.

- These disorders include
  - Postoperative Ileus (POI)

Commonly used agents – bethanechol, neostigmine

3. Genitourinary tract

- Urinary retention (atonic bladder)
  - Postoperative or postpartum
  - Neurogenic bladder (secondary to spinal cord injury or disease)

Commonly used agents – bethanechol, neostigmine

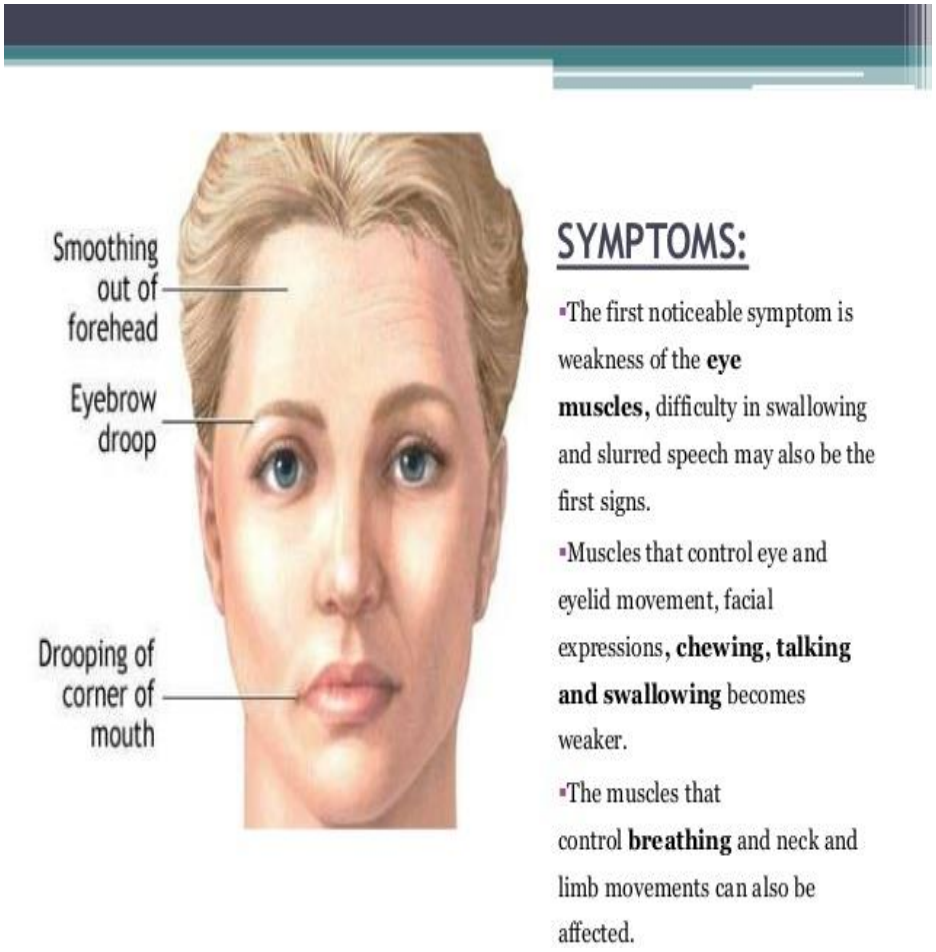
4. NMJ : Myasthenia gravis

- A chronic autoimmune neuromuscular disease that causes weakness in the skeletal muscles, which are responsible for breathing and moving parts of the body, including the arms and legs.

- Frequent findings are ptosis, diplopia, difficulty in speaking and swallowing, and extremity weakness.

- Severe disease may affect all the muscles, including those necessary for respiration.

cont'd …



- Cholinesterase inhibitors are extremely valuable as therapy for myasthenia gravis.
- Neostigmine is a drug of choice.
- Physostigmine – for long term therapy.

Tensilon test

- Edrophonium is used as a diagnostic test for myasthenia.



Tensilon test: Before (left); After (right)

Anticholinergic drugs

- Anticholinergic drugs can be classified into two based on their specific receptor affinities.

1. Muscarinic receptor blockers
2. Nicotinic receptor blockers – rarely used
   - Ganglion blockers – previously for hypertension
   - NMJ blockers – during surgery

Muscarinic receptor blockers

- Atropine (hyoscamine)(tertiary) – prototype
- Scopolamine (hyoscine) (tertiary)
- Propantheline (quaternary) – PUD
- Glycopyrrolate (quaternary)– COPD
- Tropicamide (tertiary) – eye
- Dicyclomine (tertiary) – PUD
- Tiotropium (quaternary) – asthma
- Benztropine (tertiary) - PD

Tertiary – Lipophilic
Quaternary – Polar

Pharmacokinetics (PK)

PK parameters	Tertiary	Quaternary
Absorption	Well absorbed from the gut & conjunctival membranes	Only 10–30% of a dose absorbed after oral administration
Distribution	Widely distributed in the body	Poorly taken up by the brain.
Metabolism	Hepatic	Hepatic
Excretion	Comparable	Comparable

Organ system effects

- CNS
  - In the doses usually used, atropine has minimal stimulant effects on the CNS.
  - Scopolamine has more marked central effects producing drowsiness when given in recommended dosages
  - In toxic doses, scopolamine, and to a lesser degree atropine, can cause excitement, agitation, hallucinations, and coma.

Cont'd …

- Eye
  - Mydriasis (dilated pupil)
  - Cycloplegia (inability to focus for near vision)

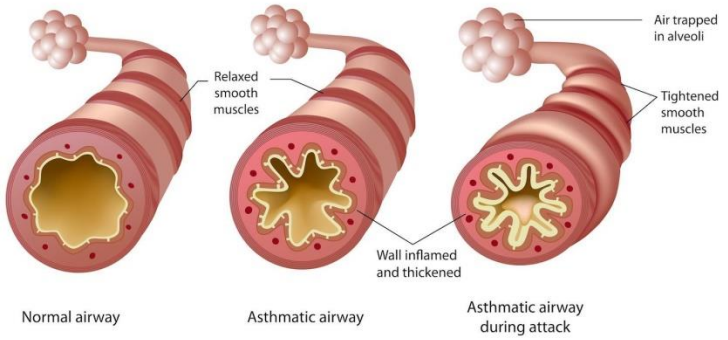




- CVS
  - Tachycardia (moderate to high dose)
  - Blocks atrial muscles
  - Ventricles are less affected at therapeutic doses.
  - Little effect on blood pressure.

Cont'd …

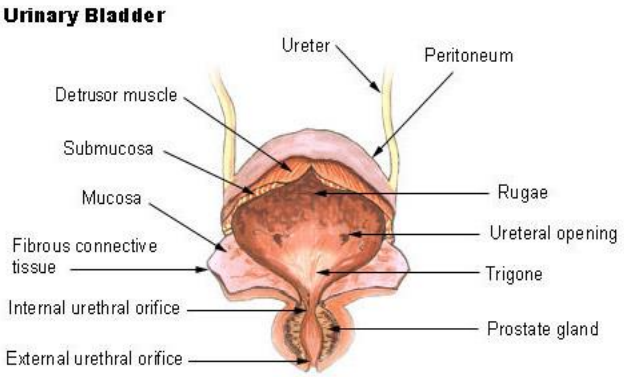
- Respiratory system
  - Bronchodilation
  - Reduced secretion
- GIT
  - Dry mouth
  - Gastric secretion is blocked less effectively.

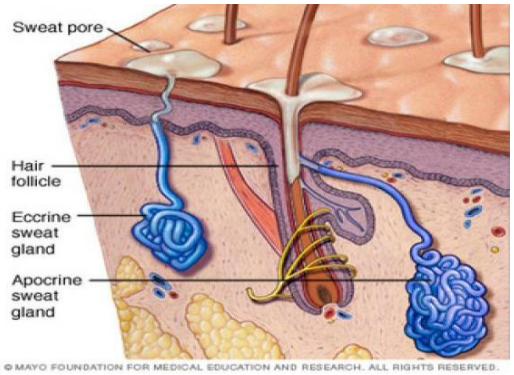




Cont'd …

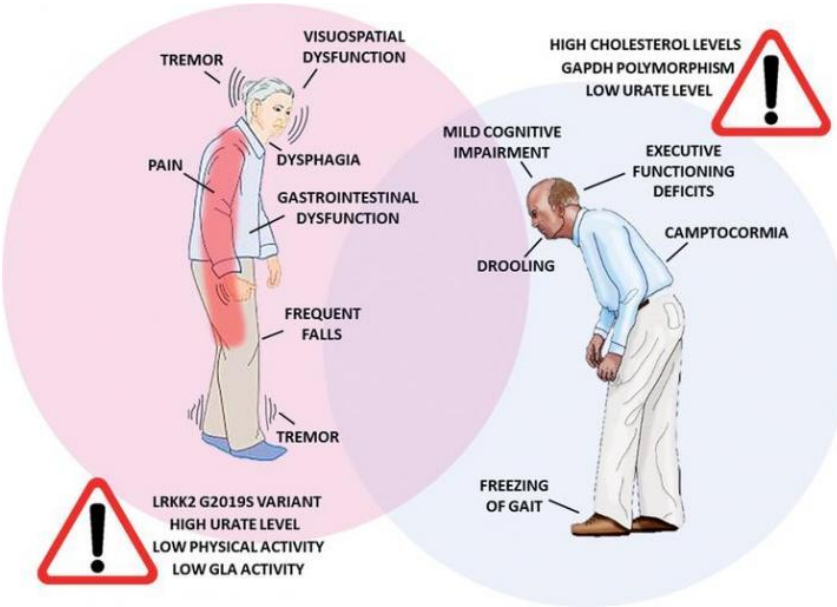
- GUT
  - Relaxes smooth muscle of the ureters and bladder wall and slows voiding.
- Sweat glands
  - Atropine suppresses thermoregulatory sweating.
  - In infants and children, this may cause "atropine fever."





Clinical pharmacology

- CNS disorders
  1. Parkinson's disease – adjunctive therapy in some patients. (Chapter - 3)



cont’d …

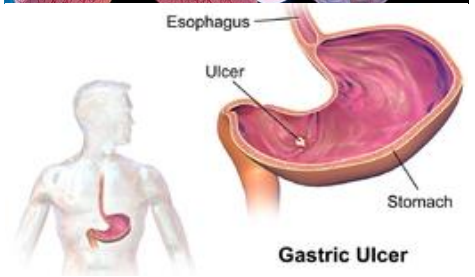
2. Motion sickness
   - Scopolamine is the drug of choice
   - Sedation and dry mouth are common.

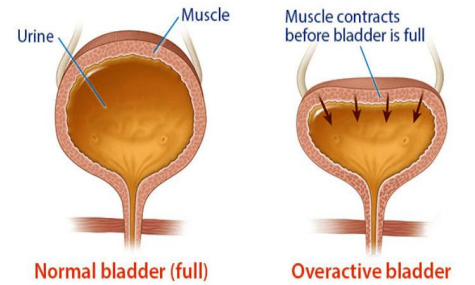


Other applications

- ✓ Respiratory disorders
  - ✓ Asthma/COPD (Ipratropium, tiotropium)
- ✓ Gastrointestinal disorders
  - – Peptic ulcer disease (propantheline)
- ✓ Urinary disorders
  - – Overactive bladder
- ✓ Cholinergic poisoning
- ✓ Hyperhidrosis



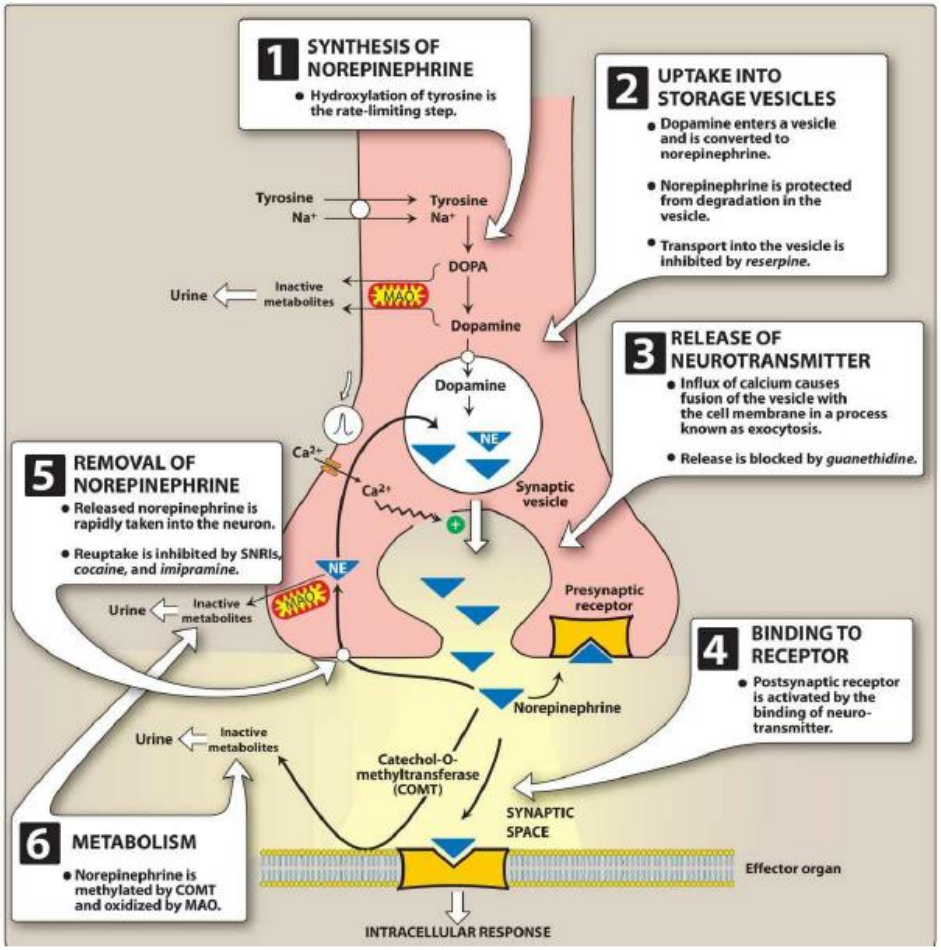






Adrenergic drugs

Neurotransmission at adrenergic neurons



Adrenergic receptors

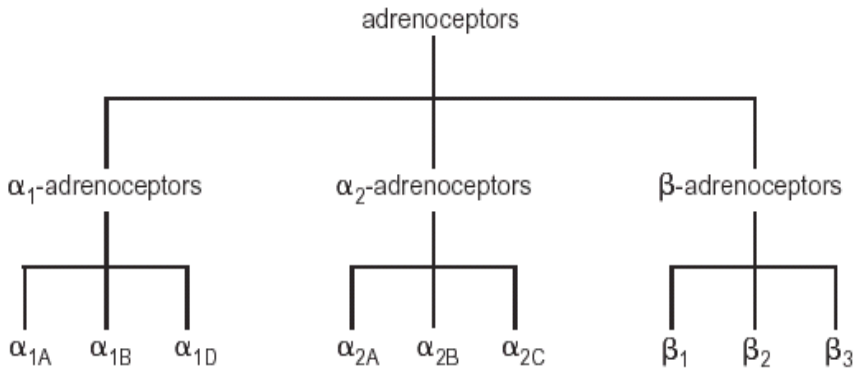
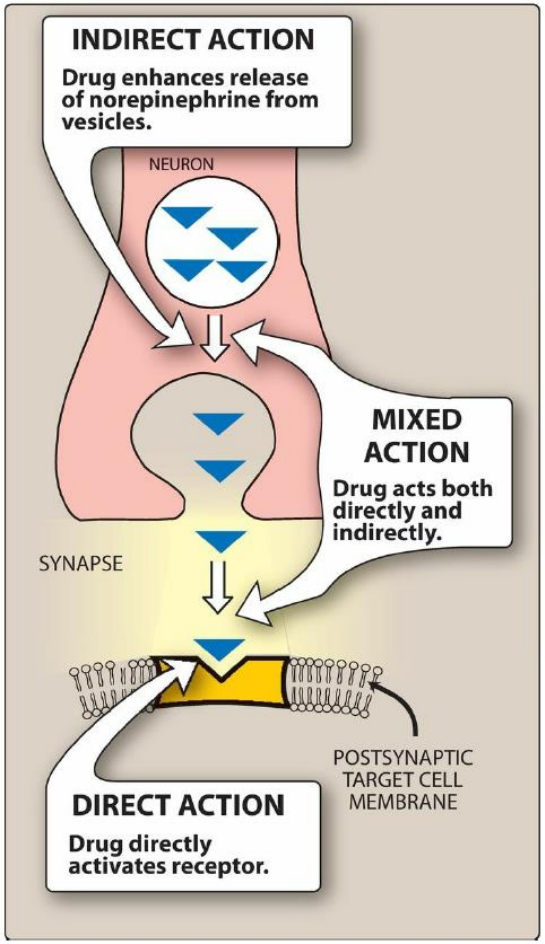


Figure. **Classification of adrenoceptors.**

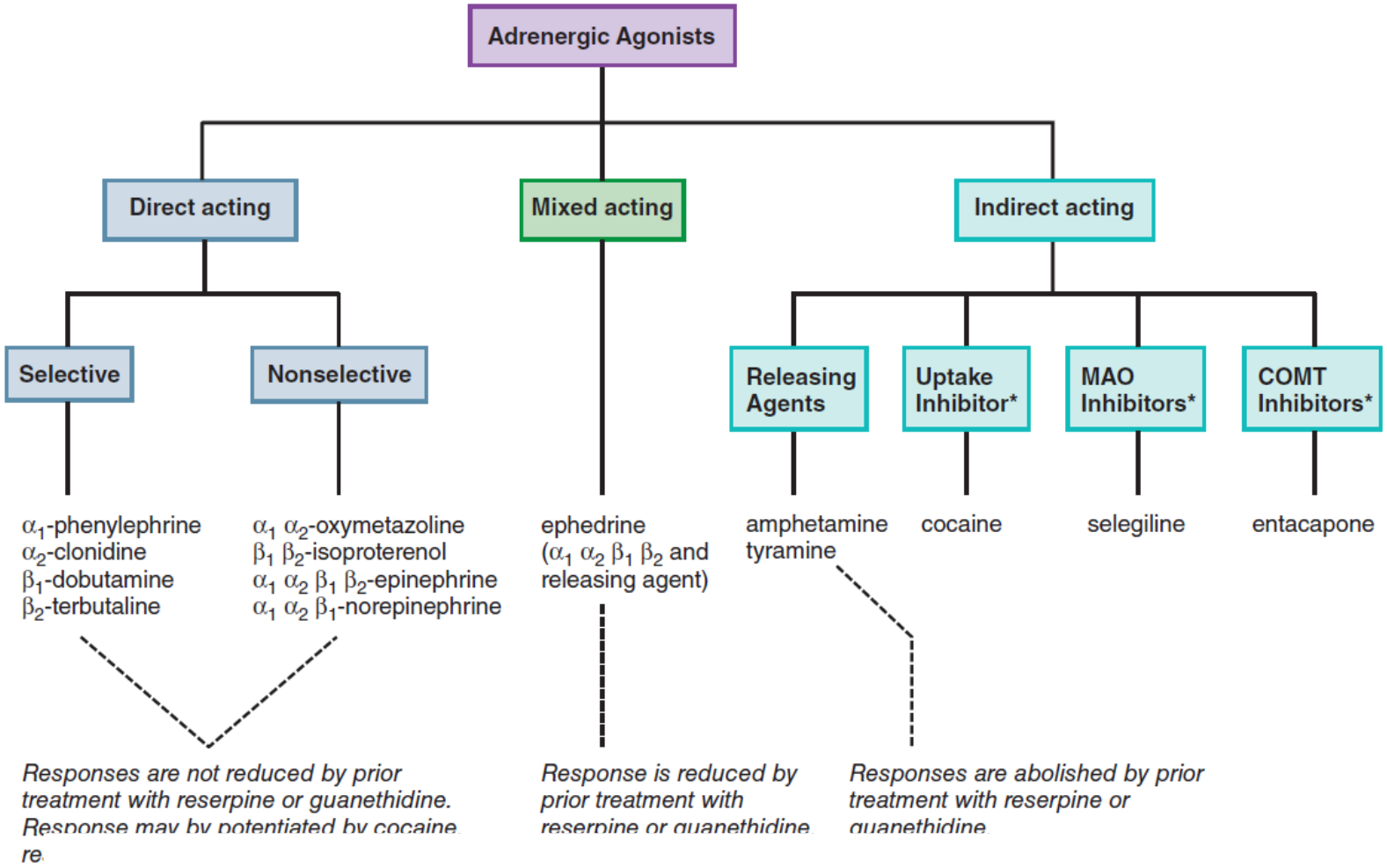
Type	Tissue	Actions
α_1	Most vascular smooth muscle (innervated)	Contraction
	Pupillary dilator muscle	Contraction (dilates pupil)
	Pilomotor smooth muscle	Erects hair
	Prostate	Contraction
	Heart	Increases force of contraction
α_2	Postsynaptic CNS neurons	Probably multiple
	Platelets	Aggregation
	Adrenergic and cholinergic nerve terminals	Inhibits transmitter release
	Some vascular smooth muscle	Contraction
	Fat cells	Inhibits lipolysis
β_1	Heart, juxtaglomerular cells	Increases force and rate of contraction; increases renin release
β_2	Respiratory, uterine, and vascular smooth muscle	Promotes smooth muscle relaxation
	Skeletal muscle	Promotes potassium uptake
	Human liver	Activates glycogenolysis
β_3	Bladder	Relaxes detrusor muscle
	Fat cells	Activates lipolysis
D_1	Smooth muscle	Dilates renal blood vessels
D_2	Nerve endings	Modulates transmitter release

Adrenergic agonists

- Classified into three as :-
  1. Direct acting
  2. Indirect acting
     - ✓ Induce release of NTs
     - ✓ Block reuptake of NTs
     - ✓ Prevent metabolism of NE
  3. Mixed action
     - ✓ Bind directly + induce release of NTs



cont'd …

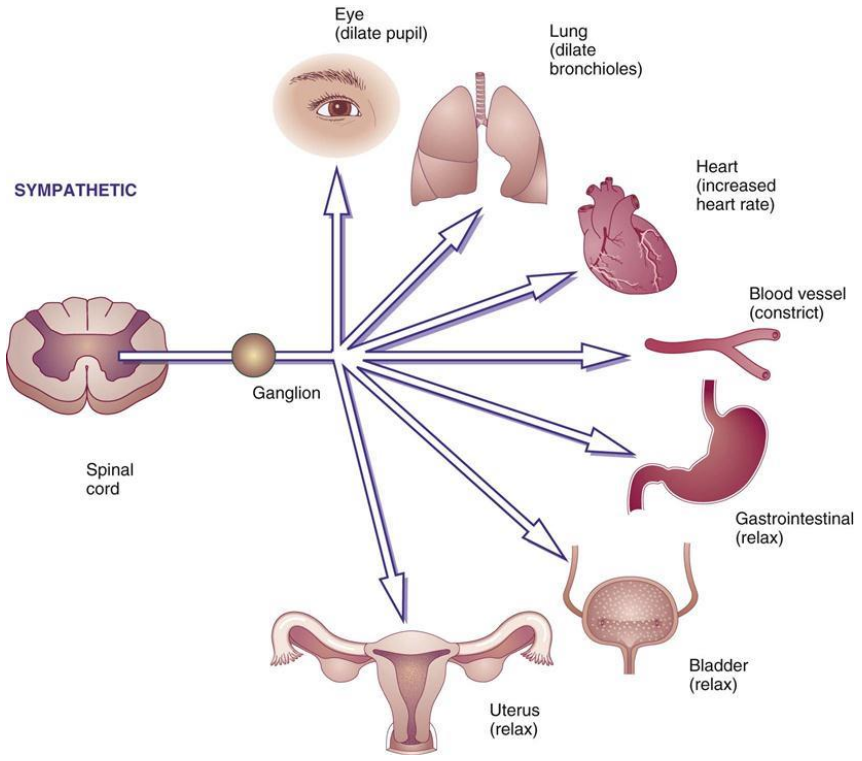


ıry, a

prototypical drug is shown. (*Not actually sympathetic drugs but produce sympathomimetic-like effects.)

Source : Goodman Gilman PHARMACOLOGICAL BASIS 13th edi

Organ system effects



Type	Tissue	Actions
α_1	Most vascular smooth muscle (innervated)	Contraction
	Pupillary dilator muscle	Contraction (dilates pupil)
	Pilomotor smooth muscle	Erects hair
	Prostate	Contraction
	Heart	Increases force of contraction
α_2	Postsynaptic CNS neurons	Probably multiple
	Platelets	Aggregation
	Adrenergic and cholinergic nerve terminals	Inhibits transmitter release
	Some vascular smooth muscle	Contraction
	Fat cells	Inhibits lipolysis
β_1	Heart, juxtaglomerular cells	Increases force and rate of contraction; increases renin release
β_2	Respiratory, uterine, and vascular smooth muscle	Promotes smooth muscle relaxation
	Skeletal muscle	Promotes potassium uptake
	Human liver	Activates glycogenolysis
β_3	Bladder	Relaxes detrusor muscle
	Fat cells	Activates lipolysis
D_1	Smooth muscle	Dilates renal blood vessels
D_2	Nerve endings	Modulates transmitter release

Specific sympathomimetic drugs

- Endogenous catecholamines
  - Epinephrine (adrenaline)
  - Norepinephrine
  - Dopamine
  - Isoproterenol (synthetic catecholamine)

Endogenous

- Epinephrine (adrenaline) (α_1, α_2, β_1, β_2)
  - A potent vasoconstrictor (α_1)and cardiac stimulant(β_1).
  - Leads to dilation of some vessels (such as skeletal muscle blood vessels) (β_2).

- Norepinephrine (noradrenaline) (α_1, α_2, β_1)
  - Activate β1 receptors with similar potency as epinephrine
  - Little effect on β2 receptors.

Direct acting sympathomimetics

- Phenylephrine (pure α1 agonist)
  - An effective mydriatic and decongestant and can be used to raise the blood pressure.

- Clonidine, methyldopa (Selective α2 agonists)
  - Decrease blood pressure through actions in the CNS that reduce sympathetic tone

cont'd …

- Oxymetazoline (α_1 , α_2)
  - Used as a topical decongestant
  - When taken in large doses, it may cause hypotension

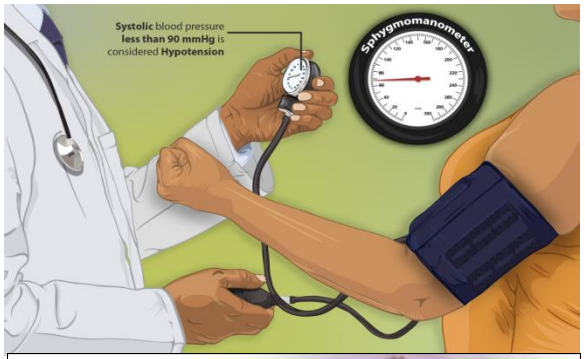
- Isoproterenol (β_1, β_2)
  - Has positive chronotropic and inotropic actions (β_1), it is a potent vasodilator (β_2) as well.

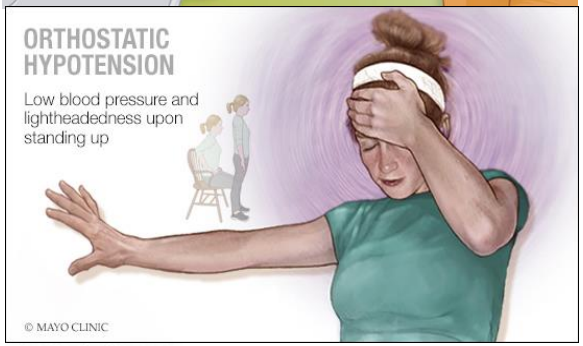
Mixed action sympathomimetics

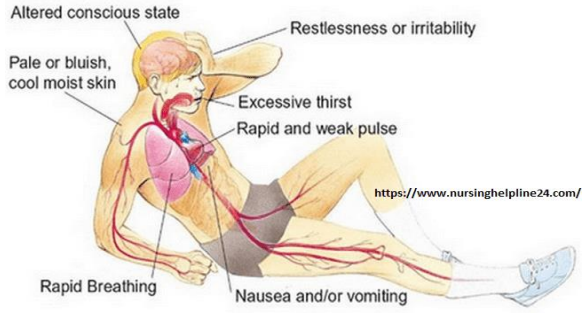
- Ephedrine (α_1, α_2, β_1, β_2 + releasing agent)
  - A non catecholamine and thus has high bioavailability and a relatively long duration of action.
  - Because it gains access to the CNS, it is a mild stimulant.

Therapeutic uses

- Cardiovascular applications
  - Treatment of acute hypotension
  - Chronic orthostatic hypotension
  - Shock
  - Cardiac applications (CST)
  - Inducing local vasoconstriction





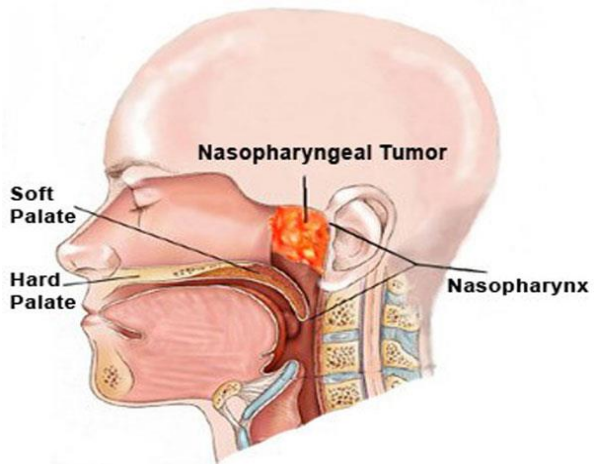




Cardiac stress test

Inducing local vasoconstriction

- Reduction of local/regional blood flow is desirable for
  - Achieving hemostasis during surgery
  - Reducing diffusion of local anesthetics away from the site of administration
  - Reducing mucous membrane congestion.





cont'd …

- Combining α agonists with some local anesthetics :-
  - Prolongs anesthetic duration of action
  - Reduced total dose of local anesthetic
  - Reduced probability of systemic toxicity



Epinephrine, 1:200,000, is the favored agent

Pulmonary applications

- Very useful in treatment of asthma or COPD
- Beta2-selective drugs are used for this purpose and include :-
  - Albuterol
  - Salbutamol
  - Metaproterenol
  - Terbutaline



Adrenoceptor antagonists

- In clinical therapeutics,
  - α1-selective antagonists are used in primary hypertension and benign prostatic hyperplasia (BPH).
  - Beta-receptor antagonist drugs are useful in a much wider variety of clinical conditions and are firmly established in the treatment of
    - Hypertension
    - Ischemic heart disease
    - Arrhythmias
    - Endocrinologic and neurologic disorders
    - Glaucoma

Basic pharmacology of alpha blockers

- MOA
  - Alpha-receptor antagonists may be reversible or irreversible in their interaction with receptors.

Pharmacologic effects

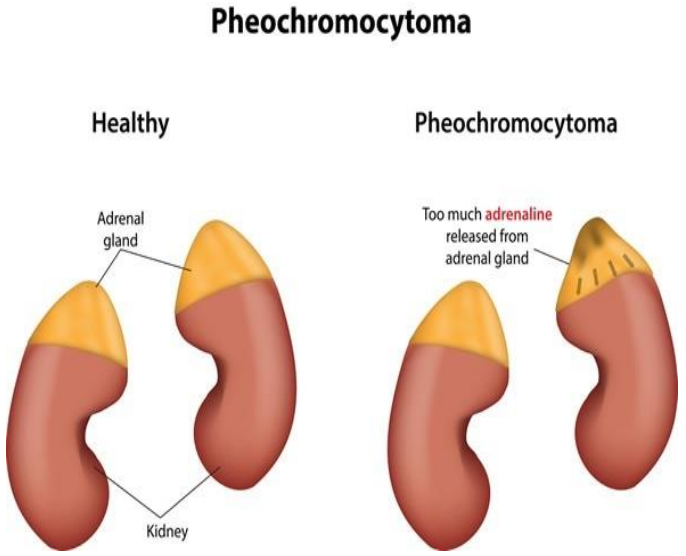
- Cardiovascular effects
  - Lower peripheral resistance and blood pressure
  - Alpha-receptor antagonists often cause orthostatic hypotension and reflex tachycardia.
- Other effects
  - Blockade of α receptors in other tissues elicits miosis (small pupils) and nasal stuffiness.
  - Alpha1 receptors are expressed in the bladder and prostate, and their blockade decreases resistance to the flow of urine.

Agents

Drugs	Receptor Affinity
Alpha antagonists	
Prazosin, terazosin, doxazosin	$\alpha_1 >>>> \alpha_2$
Phenoxybenzamine	$\alpha_1 > \alpha_2$
Phentolamine	$\alpha_1 = \alpha_2$
Yohimbine, tolazoline	$\alpha_2 >> \alpha_1$
Mixed antagonists	
Labetalol, carvedilol	$\beta_1 = \beta_2 \geq \alpha_1 > \alpha_2$
Beta antagonists	
Metoprolol, acebutolol, alprenolol, atenolol, betaxolol, celiprolol, esmolol, nebivolol	$\beta_1 >>> \beta_2$
Propranolol, carteolol, nadolol, penbutolol, pindolol, timolol	$\beta_1 = \beta_2$
Butoxamine	$\beta_2 >>> \beta_1$

Clinical pharmacology

- Pheochromocytoma
- Chronic hypertension
- Peripheral vascular disease
- Urinary obstruction (BPH)
- Hypertensive emergencies
- Erectile dysfunction



Basic pharmacology of beta blockers

- Beta blockers share the common feature of antagonizing the effects of catecholamines at β adrenoceptors.
- They differ in their relative affinities for β1 and β2 receptors.
- Some have a higher affinity for β1 than for β2 receptors, and this selectivity may have important clinical implications.

Pharmacokinetics

- Absorption
  - Most of the drugs in this class are well absorbed after oral administration.
- Bioavailability
  - Propranolol undergoes extensive hepatic (first-pass) metabolism; its bioavailability is relatively low.

Distribution and clearance

- The β antagonists are rapidly distributed and have large volumes of distribution.
- Propranolol and penbutolol are quite lipophilic and readily cross the blood-brain barrier.
- Most β antagonists have half-lives in the range of 3–10 hours. (Exception – timolol $t_{1/2}$ = 10minutes)

Organ system effects

- Cardiovascular system
  - Lower blood pressure
- Respiratory system
  - Bronchoconstriction
- Eye
  - Reduce IOP

Cont'd…

Drugs	Selectivity	Partial Agonist Activity	Local Anesthetic Action	Lipid Solubility	Elimination Half-life	Approximate Bioavailability
Acebutolol	β_1	Yes	Yes	Low	3–4 hours	50
Atenolol	β_1	No	No	Low	6–9 hours	40
Betaxolol	β_1	No	Slight	Low	14–22 hours	90
Bisoprolol	β_1	No	No	Low	9–12 hours	80
Carteolol	None	Yes	No	Low	6 hours	85
Carvedilol[1]	None	No	No	Moderate	7–10 hours	25–35
Celiprolol	β_1	Yes	No	Low	4–5 hours	70
Esmolol	β_1	No	No	Low	10 minutes	0
Labetalol[1]	None	Yes	Yes	Low	5 hours	30
Metoprolol	β_1	No	Yes	Moderate	3–4 hours	50
Nadolol	None	No	No	Low	14–24 hours	33
Nebivolol	β_1	?[2]	No	Low	11–30 hours	12–96
Penbutolol	None	Yes	No	High	5 hours	>90
Pindolol	None	Yes	Yes	Moderate	3–4 hours	90
Propranolol	None	No	Yes	High	3.5–6 hours	30[3]
Sotalol	None	No	No	Low	12 hours	90
Timolol	None	No	No	Moderate	4–5 hours	50

[1]Carvedilol and labetalol also cause α_1-adrenoceptor blockade.

[2]β_3 agonist.

[3]Bioavailability is dose-dependent.

Clinical pharmacology of beta blockers

- Hypertension
- Ischemic Heart Disease
- Cardiac Arrhythmias
- Heart Failure
- Glaucoma
- Hyperthyroidism
- Migraine headache

lol indeed!


LAUGHTER
IS THE BEST MEDICINE
And the Beta Blockers know it!
Atenolol
LOL
Metoprolol
Propranolol

THANK YOU

Good morning

- Explain about **Tensilon** test
- Points to be included in your answer
  I. The drug used for this purpose (1 point)
  II. The drug's mechanism of action (1 point)
  III. The procedure (3 points)